

Biogenic Hempcrete-Bamboo Composite with Enhanced Mechanical Performance

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Abstract

This study presents the development and mechanical characterization of new biogenic composite, termed *Plentify*, consisting of hempcrete with porous Moso bamboo (MB) fibers. Designed to support zero-mile construction strategies, *Plentify* aims to enhance mechanical performance while maintaining ecological sustainability. Uniaxial compression tests were performed on samples with 0%, 5%, 10%, and 25% bamboo fiber content. Results indicated that 10% fiber inclusion yielded the highest compressive strength at 3.19 MPa—a 30.2% improvement over the control—as well as peak stiffness, energy absorption, and ductility. The modulus of elasticity at this optimal mix reached 421 MPa, and energy absorption increased by 48.7% compared to the baseline. Although *Plentify* does not meet structural standards for load-bearing applications, its improved mechanical resilience makes it well suited for non-structural components in sustainable construction systems.

1 Introduction

This study presents *Plentify*, a biogenic material composite designed to enhance the structural performance and ecological benefits of traditional hempcrete. The composite integrates the woody core of the hemp plant (*Cannabis sativa*) with porous fibers of Moso bamboo (*Phyllostachys edulis*), and a combination of hydrated and hydraulic lime. This formulation was developed to address the mechanical limitations of conventional hempcrete, which is typically restricted to non-load-bearing applications such as insulation.

Plentify was developed from an architectural concept promoting zero-mile material sourcing—growing, harvesting, and processing materials on-site. While this framework guided its design, this paper focuses on testing *Plentify's* mechanical performance, particularly compressive strength, stiffness, and failure modes, using controlled lab tests with varying bamboo fiber content.



Figure 1: Conceptual Rendering of Construction Methodology.

Preliminary findings indicate that the inclusion of porous bamboo fibers enhances both the strength and the ductility of the reinforced concrete¹ Given the growing demand for sustainable construction materials that contribute to reduced carbon emissions, *Plentify* presents a promising alternative combining natural carbon sequestration, low energy consumption, and structural adaptability.

¹Jawad Ahmad, Zhiguang Zhou, and Ahmed Farouk Deifalla, Structural Properties of Concrete Reinforced with Bamboo Fibers: A Review, *Journal of Materials Research and Technology* 24 (2023): 844–865, <https://doi.org/10.1016/j.jmrt.2023.03.038>.

The following sections outline related research, the composite’s preparation and formulation, the testing methodology of the samples, results of testing samples, and a discussion on the speculative architectural implications in the proposed construction methodology as it changes our perception of real estate development.

2 Related Work

2.1 Sustainability, Carbon Sequestration, and Lifecycle Performance

The sustainability of hempcrete is well supported by lifecycle analysis models. Arehart et al. (2020) quantify both the biogenic carbon stored in hemp shiv and the post-curing carbonation of lime-based binders, concluding that hempcrete can achieve net-negative carbon emissions over its service life.²

A more recent state-of-the-art review by Asghari and Memari (2024) emphasizes the dual environmental and structural advantages of hempcrete. It proposes compressive strength enhancement strategies aimed at transitioning hempcrete from a low-load material to a structurally competent one—thus expanding its use in ecological architecture.³

²Arehart, Jay H., William S. Nelson, and Wil V. Srubar III. “On the Theoretical Carbon Storage and Carbon Sequestration Potential of Hempcrete.” 2020.

³Asghari, Nima, and A.M. Memari. “State of the Art Review of Attributes and Mechanical Properties of Hempcrete.” 2024.

2.2 Mechanical Performance of Natural Fiber Composites

The structural viability of natural fiber-reinforced cementitious composites has been increasingly documented. A 2022 review published in *Materials Today: Proceedings* demonstrates that fibers such as hemp, kenaf, and bamboo enhance mechanical properties substantially: compressive strength increased by 13% to 22%, tensile strength by up to 38%, and flexural strength by 23%, depending on fiber type and mix proportions. These enhancements contribute to higher toughness and crack resistance while reducing the carbon footprint of traditional concrete systems.⁴

Pushing this further, a 2025 study on hybridization effects of hemp, banana, and bamboo fibers reveals that multiscale reinforcement can lead to synergistic improvements in fracture resistance, vibrational damping, and energy absorption. In this study, the hybrid fiber-reinforced composites exhibited up to a 52% increase in fracture energy, a 45% rise in impact energy absorption, and a 32% improvement in vibrational damping capacity relative to control samples with individual fibers only. These findings confirm the mechanical superiority and functional resilience of engineered hybrid composites, validating their promise for use in load-bearing or dynamically stressed architectural elements.⁵

⁴“Jamal A. Abdalla, Blessen Skariah Thomas, and Rami A. Hawileh, “Use of Hemp, Kenaf and Bamboo Natural Fiber in Cement-Based Concrete,” *Materials Today: Proceedings*, 2022.

⁵M.A. Umarfarooq et al., “Synergetic Effects of Hemp-Banana-Bamboo Hybridization on Mechanical, Fracture and Dynamic Properties of Cellulose-Rich Natural Fiber Composites,” *International Journal of Biological Macromolecules* 320, part 3 (2025)

2.3 Synthesis

The breadth of current research confirms that natural and hybrid fiber-reinforced composites yield measurable improvements in the structural performance of cementitious and lime-based systems. Yet, while hemp, kenaf, banana, and bamboo fibers have been studied independently and in hybrid combinations within concrete matrices, there remains scant experimental data on their integration into hempcrete-based composites specifically formulated for enhanced compressive behavior.

This study will attempt at addressing the gap through the formulation and testing of *Plentify*. Rather than speculate on functional applications or broader architectural integrations, the scop here is deliberately focused: to quantify how a biogenic lime-hemp matrix responds to a bamboo fiber inclusion using standardized lab testing. This approach allows the experimental findings to directly extend-or potentially challenge- the expectation established by the prevailing body of work on natural fiber composites.

3 Methodology

We will focus on the assessment and optimization of the biogenic material composite, without engaging in the cultivation of constituent materials or the construction of planter cells—both of which are reserved for future investigations. The primary objective is to evaluate whether *Plentify* exhibits improved performance over traditional hempcrete and to determine its optimal fiber composition. The materials used include fibers from the hemp plant (*Cannabis sativa*, HP) and Moso bamboo (*Phyllostachys edulis*, MB), combined with a blend of hydrated and hydraulic lime as binders.

3.1 Sample Preparation

To investigate performance variations, mix ratios of HP and MB fibers were systematically adjusted. MB fiber content was varied from 5%, 10%, 25% to optimize structural behavior. All mixtures were formed into standardized blocks and subjected to a controlled curing process to ensure replicability.

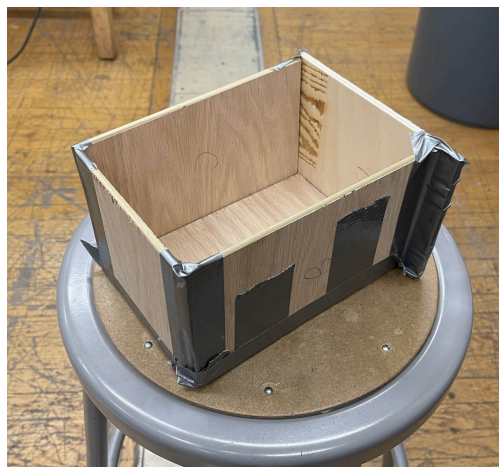


Figure 2: Plywood Mold for Sample Pouring.

The test samples were shaped into blocks measuring 152 mm $\times$ 101 mm $\times$ 101 mm (approximately 6" $\times$ 4" $\times$ 4"), dimensions suitable for comparative mechanical testing. Each sample used an estimated 16 oz of composite mixture. The base mixture followed a ratio of 1:1.5:2 for HP hurd to binder (lime binder) to water. For the control block, the materials were measured as follows:

Table 1: Control Sample Material Quantities

Material	Quantity
Hemp shiv (HP hurd)	45 grams
Lime Binder	67.5 grams
Water	90 grams (90 ml)

This combination yields a total volume of approximately 16 oz prior to molding.

Safety Note: Due to the caustic nature of lime, appropriate safety precautions were followed. Personal protective equipment—including gloves, eye protection, and a respirator mask—was used at all times. In case of exposure, affected areas were rinsed thoroughly with mild soap and water.

3.2 Mixing Procedure

The mixing process began with ensuring all materials—particularly the hemp hurd—were thoroughly dried and free from impurities. Next, the hemp hurd was blended with the lime-pozzolan binder in a clean plastic container until a uniform dry mix was achieved. Water was then added gradually while continuously stirring, with care taken to reach a moldable, non-slumping consistency. Once mixed, the material was pressed firmly into 6-inch molds to ensure uniform compaction and eliminate air gaps. The molded samples were left in place for 24 to 48 hours before being removed and air-cured for a minimum of four weeks. A general guideline of one week of curing per 1.5 inches of depth was followed to ensure adequate hardening.



Figure 3: Dry Mixing in Plastic Container.

3.3 Sample Composition Adjustments

To evaluate the impact of MB fiber integration, three modified samples were prepared in addition to the control, with the total net mixture weight maintained at approximately 16 oz. Bamboo fiber content was increased progressively:

Table 2: Bamboo Fiber Content in Composite Samples

Sample ID	Bamboo Fiber Content
A0	Control (0% bamboo fibers)
A5	5% bamboo fibers
A10	10% bamboo fibers
A25	25% bamboo fibers

These controlled variations enabled comparative analysis of mechanical performance across different fiber compositions.

Note: The mixing and molding procedures for all samples follow a consistent protocol, with the only variation occurring during the dry ingredient phase—specifically in the incremental addition of bamboo fibers. By systematically increasing the bamboo fiber content, this study aims to identify the optimal inclusion rate that enhances the structural and insulation performance of *Plentify*. Maintaining a consistent total sample mass across all formulations ensures accurate and controlled comparative analysis.

Table 3: Dry Ingredient Composition of Composite Samples

Sample	Hemp Shiv (g)	Bamboo Fibers (g)	Lime Binder (g)	Total Dry Weight (g)
A0	45	0.00	67.5	90.0
A5	45	12.50	67.5	125.0
A10	45	19.88	67.5	132.38
A25	45	37.50	67.5	150.0

Table 4: Water Content by Sample

Sample	Water (ml)
A0	90.00
A5	100.04
A10	105.91
A25	119.93

4 Testing Methodology

To evaluate the mechanical performance of the *Plentify* biogenic composite, a series of uniaxial compression tests were conducted. These tests aimed to determine whether the inclusion of Moso bamboo (MB) fibers improves the compressive strength and stiffness of traditional hempcrete. All procedures were adapted to accommodate the composite's geometry and adhered to ASTM C109 guidelines, which govern compressive strength testing for hydraulic cement mortars using cube-shaped specimens.

Compression testing was performed using an MTS Insight Electromechanical Testing System with a 100 kN load capacity. The test setup included dual hot-rolled steel platens, each measuring approximately 0.75 inches by 8 inches by 8 inches and weighing 12.3 lbs. Each specimen was carefully centered on the lower platen to ensure axial load application. Prior to loading, precise alignment was verified to eliminate off-axis deviations that could compromise test accuracy.

A constant crosshead displacement rate was maintained throughout testing, targeting a nominal stress application rate between 0.2 and 0.3 MPa per second. A small initial preload was applied to seat the specimen; however, this value was excluded from the final data analysis. Force and displacement data were collected using MTS proprietary software. Where applicable, additional strain data were recorded via video extensometry to provide detailed insight into deformation behavior.

4.1 Variables Measured

The mechanical behavior of the composite was characterized by systematically recording and analyzing a series of variables. Primary measurements included the load (N), representing the total axial force applied during compression, and displacement (mm), which tracked the vertical deformation of the specimen under load and was used to calculate strain. From these, several derived mechanical properties were calculated: stress (MPa), obtained by dividing the applied load by the cross-sectional area ($152 \text{ mm} \times 101 \text{ mm}$), indicated internal resistance; strain (%), derived from displacement, reflected the relative deformation compared to the original specimen height; compressive strength (MPa) measured the maximum stress the specimen withstood before failure; and the modulus of elasticity (MPa), calculated from the slope of the linear-elastic region of the stress–strain curve, indicated the material’s stiffness. Supplemental metrics included average pressure per inch (psi/in), which represented the load distribution per unit height and was useful for scaling and design applications. Additionally, qualitative observations such as the failure mode—whether crushing, splitting, or barreling—were recorded through visual inspection and photography to provide further insight into the material’s structural behavior.

These variables were selected to enable both quantitative comparison and qualitative interpretation of the composite’s mechanical behavior. The control specimen—composed of conventional hempcrete without bamboo fibers—served as the baseline for all analyses. Comparative results from experimental groups incorporating 5%, 10%, and 25% MB fibers were used to assess trends in compressive strength, stiffness, and failure characteristics. This incremental analysis was instrumental in identifying the optimal fiber inclusion rate and evaluating the structural viability of *Plentify*.

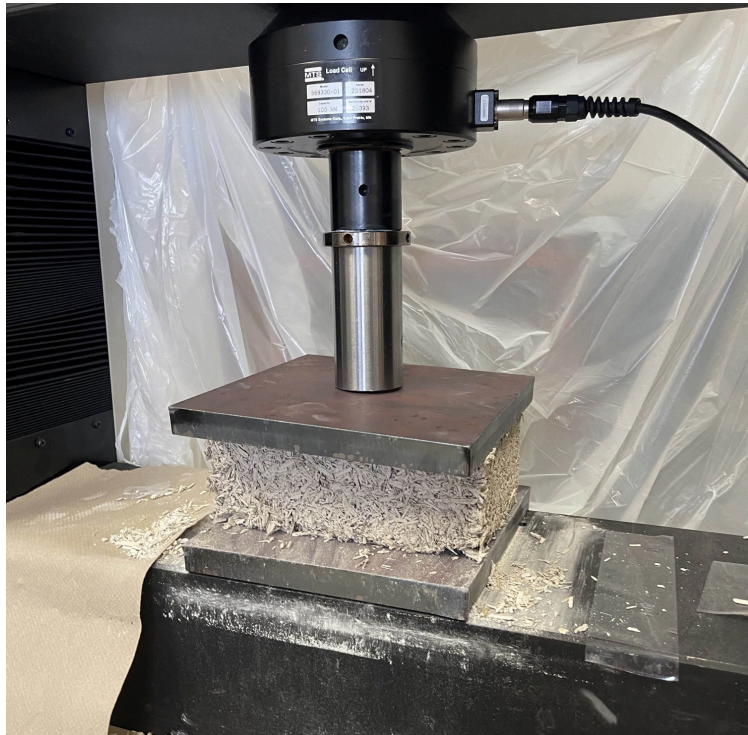


Figure 4: Beginning of Test.



Figure 5: End of Test.

5 Results

Compression tests were performed on four distinct specimens of the *Plentify* composite to evaluate their mechanical behavior under axial loading. The specimens varied by the inclusion rate of porous Moso bamboo fibers: A0 (control, 0%), A5 (5%), A10 (10%), and A25 (25%). For each specimen, the parameters recorded included time (s), load (N), extension (mm), stress (MPa), and strain (%). Stress was calculated by dividing the load by the constant cross-sectional area ($22,500 \text{ mm}^2$), and strain was derived from the extension relative to the original height (150 mm).

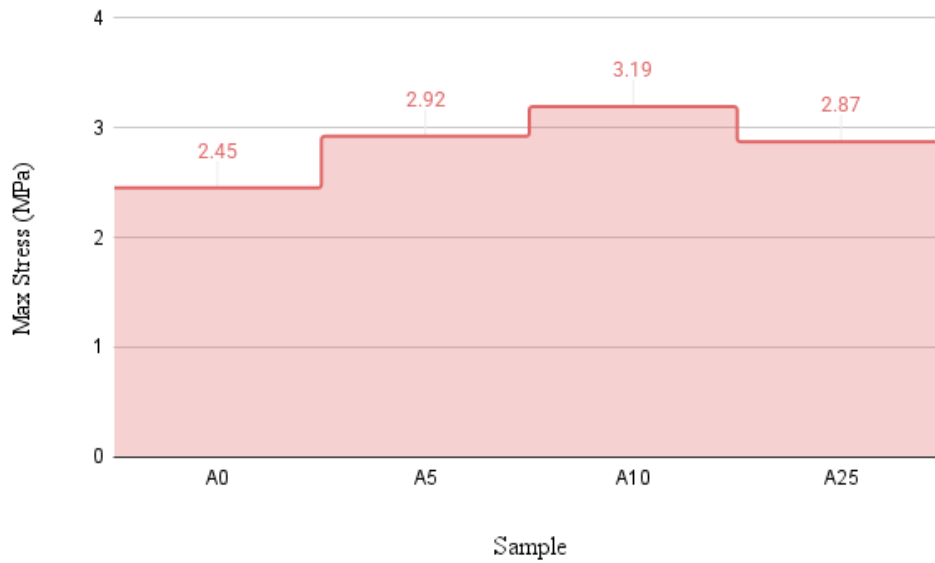


Figure 6: Maximum Compressive Strength.

A10 achieved the highest compressive strength of 3.19 MPa, reflecting a 30.2% increase over A0 (2.45 MPa).

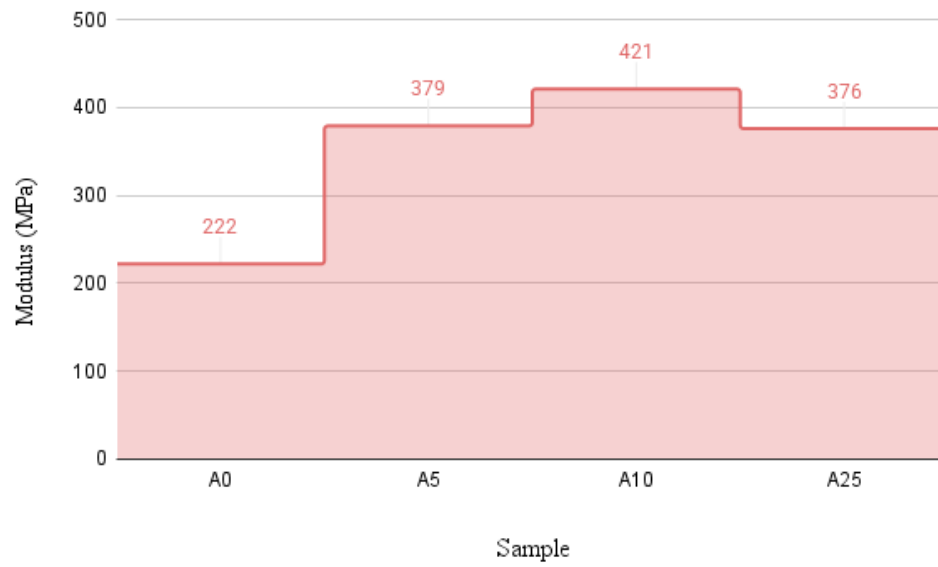


Figure 7: Modulus of Elasticity.

A10 had the highest stiffness, with a modulus of elasticity of 421 MPa—an 89.6% increase over A0's modulus of 222 MPa.

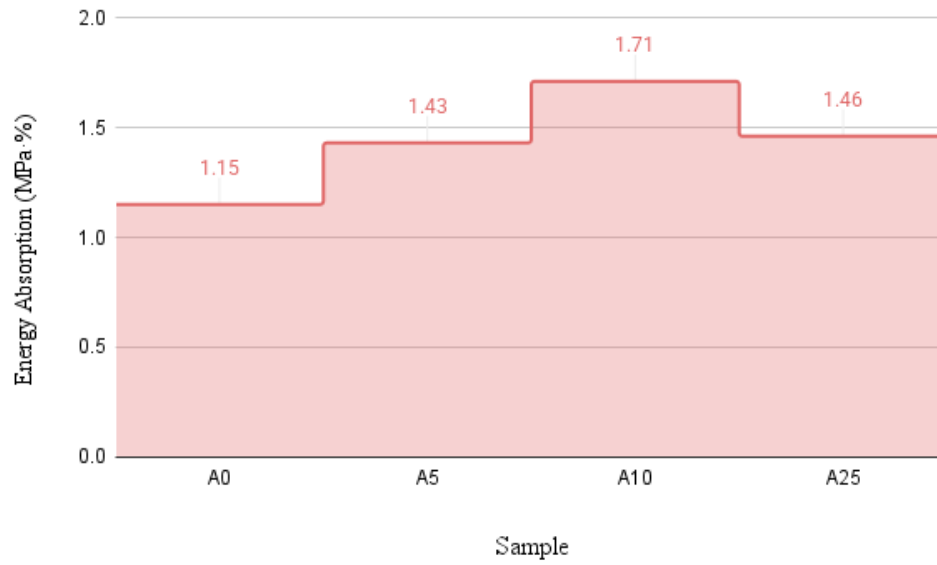


Figure 8: Energy Absorption.

A10 absorbed 1.71 MPa·% in energy, representing a 48.7% increase in toughness compared to A0's 1.15 MPa·%.

Table 5: Failure Strain

Sample	Failure Strain (%)
A0	0.83
A1	1.07
A2	1.25
A3	1.10

A10 reached a failure strain of 1.25%, a 50.6% improvement in ductility over A0's 0.83%.

5.1 Specimen Performance Summary

Sample A0 (Control, 0% Bamboo Fiber): This control sample exhibited a gradual increase in stress relative to strain, reaching a peak compressive strength of approximately 2.45 MPa at a failure strain of 0.83%. The stress-strain curve featured a short linear elastic phase followed by gradual strain accumulation prior to failure. The estimated modulus of elasticity was 222 MPa, and energy absorption up to failure was 1.15 MPa·%.

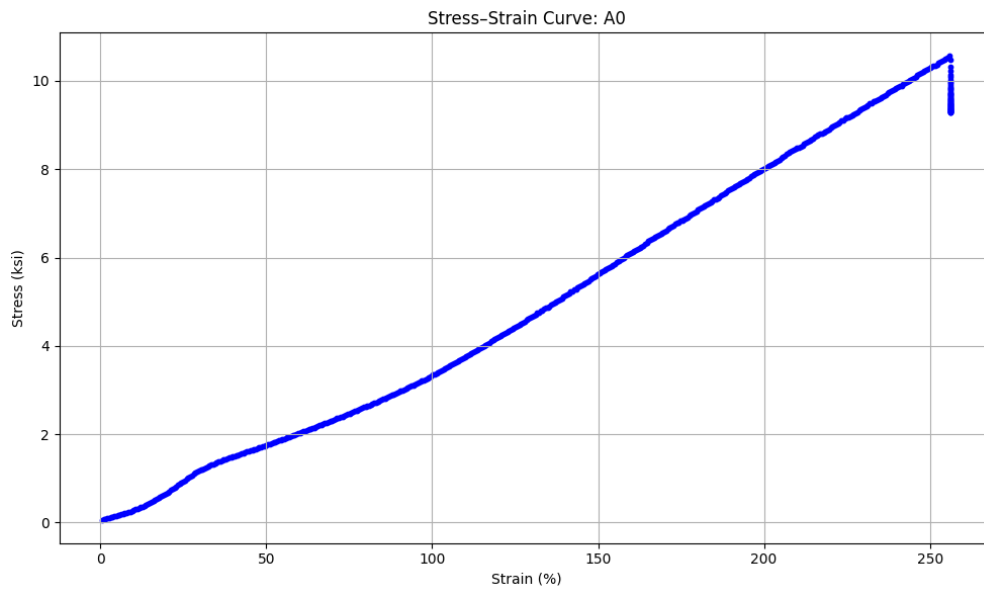


Figure 9: Stress-Strain curve for Sample A0 (0% bamboo fiber).

Sample A5 (5% Bamboo Fiber): This sample demonstrated improved mechanical response over the control. It achieved a peak compressive strength of 2.92 MPa and a failure strain of 1.07%. The modulus of elasticity increased to 379 MPa, while energy absorption rose to 1.43 MPa·%. The stress-strain curve maintained a shape similar to A0 but was elevated throughout, indicating better deformation handling and load capacity.

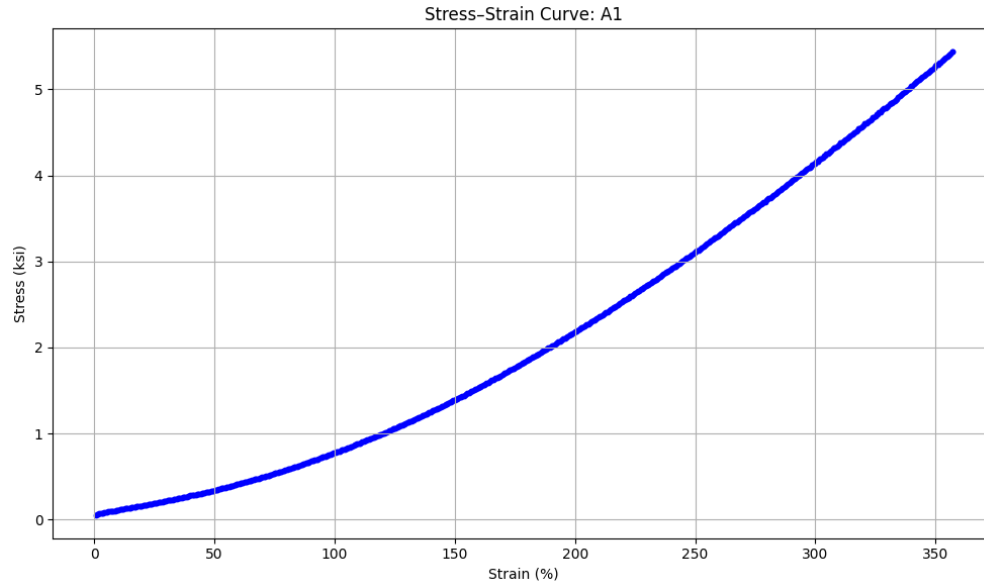


Figure 10: Stress-Strain curve for Sample A5 (5% bamboo fiber).

Sample A10 (10% Bamboo Fiber): This composition yielded the highest mechanical performance. The peak compressive strength reached 3.19 MPa, with a failure strain of 1.25%. A pronounced and extended linear elastic region was observed, and the modulus of elasticity was estimated at 421 MPa—the highest among all samples. Energy absorption reached a maximum of 1.71 MPa·%, reflecting optimal reinforcement efficiency at this fiber content.

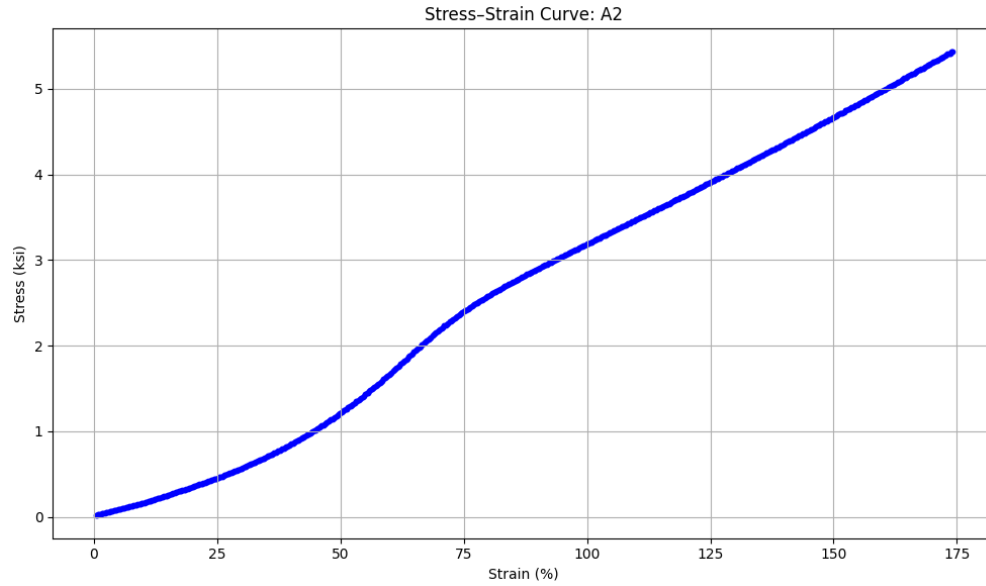


Figure 11: Stress-Strain curve for Sample A10 (10% bamboo fiber).

Sample A25 (25% Bamboo Fiber): Although outperforming the control, A25 showed slightly diminished performance compared to A10. The peak compressive strength was 2.87 MPa, with a failure strain of 1.10%. The modulus of elasticity measured 376 MPa, and energy absorption was 1.46 MPa·%. The stress-strain curve featured a shorter linear region and earlier onset of nonlinear deformation than A10.

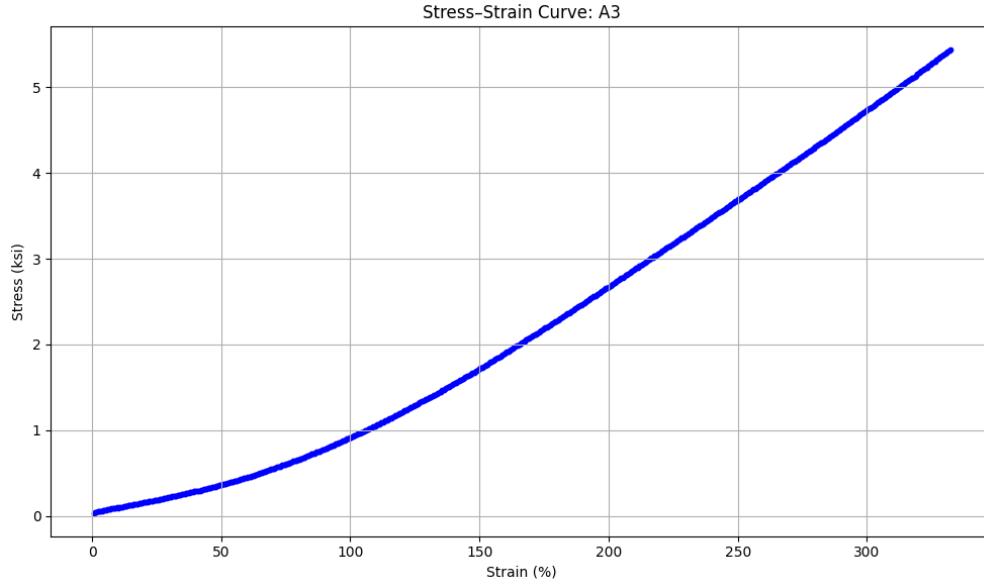


Figure 12: Stress-Strain curve for Sample A25 (25% bamboo fiber).

5.2 Trend Analysis

Mechanical performance exhibited a generally increasing trend across the samples, with compressive strength, modulus of elasticity, failure strain, and energy absorption all improving from A0 to A10. While A10 recorded the highest values across most metrics, the decline observed in A25 was relatively modest, and intermediate fiber concentrations such as 15% or 20% were not tested. As such, while the data indicate strong performance at 10% fiber content, they do not conclusively establish it as the optimum. The possibility remains that the true threshold for peak performance could lie between 10% and 25%. Future studies incorporating finer gradations of fiber content will be necessary to more accurately identify the inflection point of mechanical efficiency.

5.3 Failure Characteristics

All specimens exhibited barreling under compressive loading, confirmed through visual inspection. Post-yield deformation was observed consistently, culminating in crushing failure. Load–deformation profiles showed progressive strain accumulation under increasing load until maximum stress was reached, followed by structural collapse.



Figure 13: Sample A0 Barreling.

6 Discussion

The results have significant implications for the application of *Plentify* in architectural and sustainable design contexts. The substantial increases in compressive strength, stiffness, and energy absorption observed at 10% bamboo fiber content suggest that fiber-reinforced *Plentify* may be structurally viable for non-load-bearing components subjected to moderate mechanical stresses. Potential applications include precast panels, partition walls, and eco-integrated architectural elements where both mechanical robustness and environmental sustainability are important.

The initial intentions behind the composite’s conception was an architectural concept of on-site material growth, or zero-mile construction. In this, users would grow both plants (bamboo and hemp) within modular planters that in time would be retro-fitted into walls themselves. The proposed methodology had a slower development time frame, adjusting to varying economic levels for areas with significantly lower development.

Although the addition of bamboo fibers markedly improved the composite’s mechanical performance, compressive strengths remain below the thresholds required for load-bearing structures. For reference, conventional concrete used in structural walls typically requires strengths exceeding 17 MPa. In contrast, the highest-performing *Plentify* sample (A10) achieved 3.19 MPa—a 30.2% improvement over the control but still within the range of non-structural applications.

The observed trends align with theoretical models of fiber-reinforced composites. According to the rule-of-mixtures theory, strength and stiffness should increase with high-modulus fiber inclusion, provided interfacial bonding is adequate. This model is supported by the data, as Sample A10 exhibited peak performance, while Sample A25 (25% fiber) experienced a slight decline, likely due to poor dispersion or increased void content.

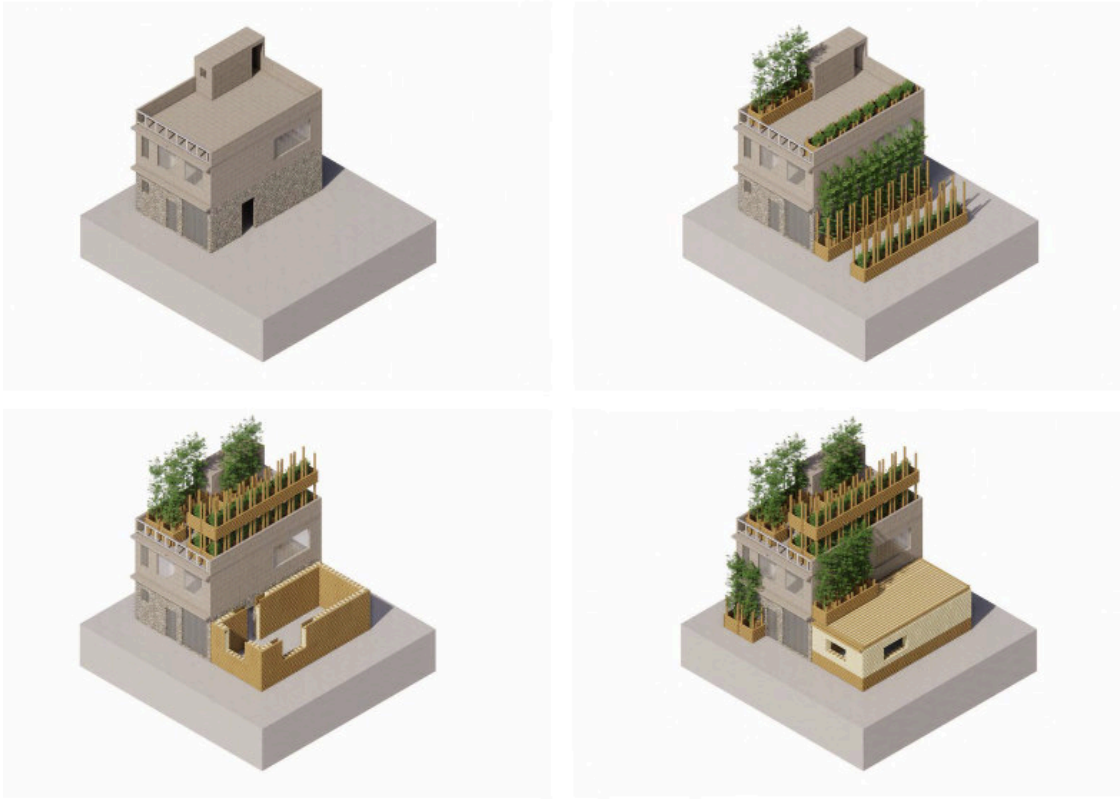


Figure 14: Conceptual Growth Methodology.

To further optimize the composite, future development should explore surface modification of the bamboo fibers (e.g., alkaline or silane treatments) to enhance matrix adhesion. Additional testing of tensile, flexural, and impact properties is recommended to provide a complete mechanical profile. Likewise, evaluating thermal and acoustic behavior would extend relevance to passive and energy-efficient design strategies.

Sample A10's 3.19 MPa strength places it above typical hempcrete performance (0.5–3.0 MPa) and 6.3% higher than the upper range of standard formulations. These findings confirm that bamboo fiber inclusion significantly enhances the structural capacity of the composite.

7 Conclusion

This research investigated the mechanical performance of a biogenic composite integrating hempcrete with porous Moso bamboo fibers, termed *Plentify*. Uniaxial compression testing revealed that moderate fiber inclusion—particularly at 10%—substantially enhances compressive strength, stiffness, and ductility. Although the composite does not yet meet structural-grade criteria for load-bearing applications, it holds promise for non-structural components in sustainable construction.

The findings contribute to broader efforts in material science to reduce the carbon footprint of the built environment. *Plentify* combines carbon sequestration, thermal regulation, and ecological sourcing, making it a viable candidate for low-impact architectural use. However, future studies are needed to strengthen fiber-matrix interaction, evaluate broader mechanical behaviors, and test environmental properties such as insulation and sound attenuation.

In summary, *Plentify* offers a compelling model for next-generation building materials. Its continued refinement may unlock its potential as a high-performance, sustainable alternative in bio-based architectural design.

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